Gas Injury Detection

Wearable Sensor Classification — Progress Report

June 5, 2026

Key Results

- **The current sensor network reliably detects blood mixtures down to 20% blood.**
- The best model, **XGBoost with feature reduction and causal temporal smoothing**, runs in real time and reaches **95.8% accuracy** on held-out sessions, with the few remaining errors confined to the moment a sample is introduced.
- **NH_3 is the dominant blood marker** — its level scales with blood concentration, consistent with ammonia release from protein breakdown in blood.

Detection Floor: Why `mix_140_10` Was Excluded

`mix_140_10` is the lowest-concentration mixture: 6.7% blood in 93.3% sweat. NH_3 is the primary discriminator between blood and sweat, and its response scales with blood content. At 6.7% the NH_3 response falls into the pure-sweat range, and two of the five `mix_140_10` sessions showed no measurable NH_3 elevation above sweat on either the peak or the sustained level.

Since the two classes overlap in the sensor reading, the model cannot separate them, and including `mix_140_10` only adds noise to the blood class, so these sessions were excluded. The next ratio up, `mix_120_30` at 20% blood, produces roughly double the sweat NH_3 and separates cleanly, which sets 20% as the lowest blood content the network can reliably detect.

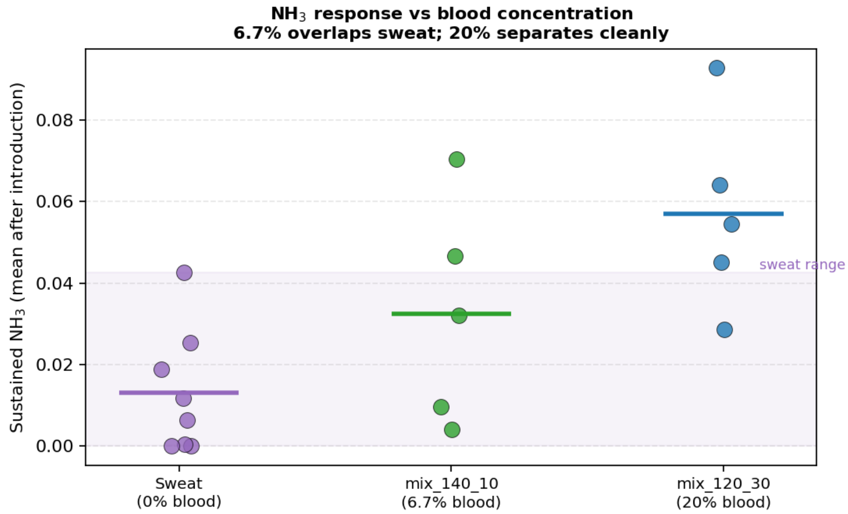

Figure 1: Sustained NH_3 per session by sample type. The 6.7% blood mixture (`mix_140_10`, green) interleaves with the pure-sweat range (shaded), while the 20% mixture (`mix_120_30`, blue) sits clearly above it. This overlap is why 6.7% samples cannot be added to the model.

Models Evaluated

Three model families were compared, chosen to span interpretability, temporal-modeling capacity, and suitability for embedded deployment.

Table 1: Model characteristics.

Model	Strengths	Limitations
Random Forest	Interpretable feature importances; robust on small tabular datasets; no input scaling required.	Requires hand-crafted window aggregates; does not learn from the raw time-series.
XGBoost	Gradient boosting corrects residual errors sequentially, outperforming RF on the same features; L1/L2 regularization in the training objective penalizes model complexity directly.	Same tabular feature dependency as RF; less interpretable.
1D CNN	Learns temporal patterns directly from the raw signal; no manual feature engineering.	Data-hungry; input normalization suppressed the low-magnitude NH_3 signal, so it confused sweat with blood.

The Random Forest and XGBoost operate on 60 s window aggregates (mean, std, max, slope) of the engineered feature set. The CNN consumes the raw 30 s window directly. Across held-out sessions the tree models were clearly stronger: the task is fundamentally one of signal *level* (how high NH_3 rises), which the raw feature values expose directly, whereas the CNN's per-channel scaling compressed that level away.

Dataset

Data was collected across 28 sessions using the ESP32-based sensor array (VOC, NH_3, HCHO, H_2S, EtOH, plus temperature and humidity). The 6.7% mixture (`mix_140_10`) is excluded for the reason described above. Each session is manually labeled into baseline, sweat, and blood regions.

Splits are performed at the session level so that all windows from a given session stay together, which prevents temporally correlated windows from leaking between training and test. 24 sessions are used for training and 4 are held out as unseen test data. Non-overlapping 60 s windows produce 1,683 windows in total; the training windows are class-balanced (sweat bootstrapped, baseline undersampled) before the tree models are fit.

Table 2: Dataset summary.

Property	**Count**
Total sessions	28
Sweat	8
Pure blood	10
Mixture, 50% blood (`mix_75_75`)	5
Mixture, 20% blood (`mix_120_30`)	5
Training sessions	24
Held-out test sessions	4
Total windows (60 s)	1,683
Test windows	215

Training Set Balancing

The 1,468 raw training windows are heavily imbalanced. Baseline dominates because each session spends most of its time before the sample is introduced, and sweat is the minority class. Training on this distribution would bias the model toward baseline and underrepresent sweat, so the three classes are balanced to the blood count of 442 windows.

Bootstrapping the sweat class. Each synthetic sweat window is created by drawing an existing sweat window at random (with replacement) and adding Gaussian noise to every feature, scaled to 5% of that feature's standard deviation across the sweat windows: $x_{\text{syn}} = x + \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.05 \times \text{std}(x)$. The noise keeps each synthetic window recognizably sweat while avoiding exact duplicates, which would only reweight the same points without adding variety. This adds 282 synthetic windows, taking sweat from 160 to 442.

Undersampling the baseline class. Baseline is randomly subsampled from 866 down to 442 windows.

The final balanced training set is 1,326 windows, 442 per class. Balancing is applied to the training data only; the held-out test sessions keep their natural class distribution.

Table 3: Training window counts before and after balancing.

Class	Raw windows	After balancing
Baseline	866	442 (undersampled)
Sweat	160	442 (+282 bootstrapped)
Blood	442	442 (unchanged)
Total	**1,468**	**1,326**

Model Performance

Across model families, the gradient-boosted trees were the strongest. Table 4 compares overall accuracy on the four held-out sessions. XGBoost with importance-ranked feature reduction (top 80% of features) is the best base model.

Table 4: Overall accuracy on held-out test sessions (sweat, pure blood, and 20% / 50% mixtures), before temporal smoothing.

Model	Overall accuracy	Notes
Random Forest (all features)	96.7%	200 trees, balanced classes
XGBoost (naive)	97.7%	all features, defaults
XGBoost (20% feature reduction)	**98.6%**	importance-ranked pruning
1D CNN (regularized)	89.4%	level signal lost in scaling

Best Model: XGBoost with Real-Time Smoothing

Because each 60 s window is classified on its own, the raw predictions can flicker between classes from one window to the next. Since the windows form a time series, a causal temporal smoother averages the predicted probabilities over the last 5 windows before deciding, using only current and past data so it runs in real time. This produces stable predictions suitable for live deployment.

On the held-out sessions the smoothed real-time model reaches **95.8% overall** (Table 5). The small gap from the unsmoothed 98.6% is onset lag: the smoother takes a few windows to commit after a sample is introduced. Almost all remaining errors fall in that onset window; once the response is established, predictions are correct and stable.

Table 5: Per-session accuracy of the deployed model (XGBoost, 20% feature reduction, causal real-time smoothing).

Session	**Type**	**Accuracy**
`blood_9`	Pure blood	97.0%
`sweat_7`	Pure sweat	96.6%
`mix_75_75_3`	Mixture, 50% blood	95.7%
`mix_120_30_4`	Mixture, 20% blood	93.0%
Overall	(215 windows)	**95.8%**

The figures below show the model's prediction over each held-out session. Green marks windows the model classified correctly, red marks errors. The error band sits at the sample introduction in every session; the long baseline and response regions are predicted correctly.

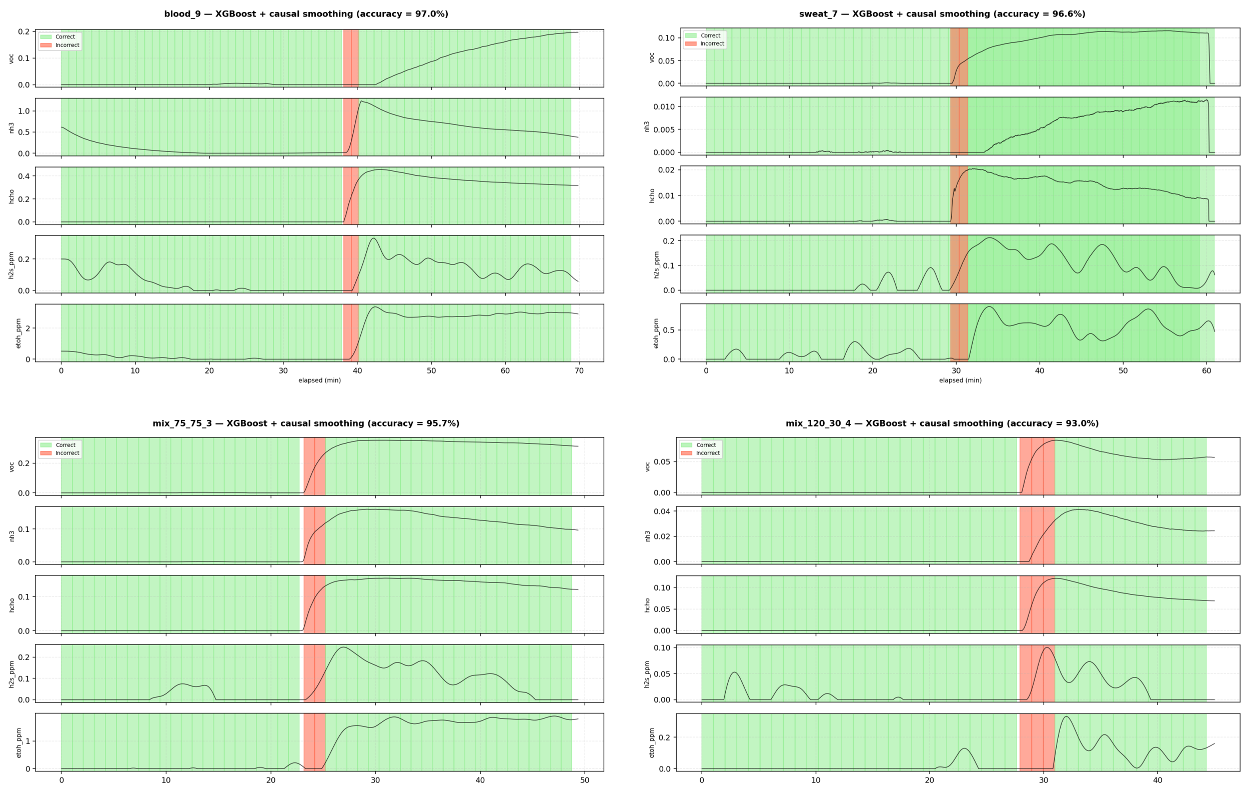

Figure 2: Real-time predictions per held-out session (green = correct, red = incorrect). Errors are confined to the onset transition when the sample is first introduced.